

Project Plan

The project plan outlines the complete development process of the Web-Based Phishing Detection System, including timelines, tasks, milestones, and deliverables.

The project is divided into well-defined phases to ensure smooth progress and effective monitoring.

Phase 1: Requirement Analysis & Planning

→ **Duration: Weeks 1–2**

Activities:

- Identification of functional and non-functional requirements
- Understanding phishing attack patterns and detection needs
- Finalization of project scope and objectives
- Selection of technologies and tools

Deliverables:

- Approved necessary documents
 - Finalized project scope and objectives
-

Phase 2: Literature Review & Research

→ **Duration: Weeks 3–4**

Activities:

- Study of existing phishing detection techniques
- Review of rule-based, blacklist, and ML-based systems
- Identification of research gaps and limitations
- Alignment of project approach with academic literature

Deliverables:

- Comprehensive Literature Review document
 - Problem justification and solution direction
-

Phase 3: System Design

→ **Duration: Weeks 5–6**

Activities:

- Design of system architecture
- Frontend UI flow design
- Backend processing workflow
- Detection logic and ML model design
- Chatbot interaction design

Deliverables:

- System Architecture Diagram
 - Module-level design description
 - Data flow and detection flow diagrams
-

Phase 4: Implementation (Development)

→ **Duration: Weeks 7–10**

Activities:

- Development of frontend using HTML, CSS, and JavaScript
- Backend development using Python (Flask)
- Implementation of rule-based phishing detection
- Integration of Machine Learning model (Random Forest)
- Chatbot implementation for user guidance

Deliverables:

- Functional web application
 - Integrated detection and chatbot modules
-

Phase 5: Testing & Validation

→ **Duration: Weeks 11–12**

Activities:

- Functional testing using safe and phishing URLs
- Validation of detection accuracy
- Performance and usability testing
- Requirement validation against SRS

Deliverables:

- Test cases and test results
 - Validated system functionality
-

Phase 6: Documentation & Mid-Year Evaluation

→ **Duration: Weeks 13–14**

Activities:

- Preparation of FYP-I documentation
- Compliance report preparation
- Minutes of Meeting documentation
- Mid-year evaluation preparation

Deliverables:

- Complete FYP-I documentation
 - Compliance report
 - Approved Mid-Year evaluation
-

Phase 7: Future Enhancements (Planned for FYP-II)

→ **Duration: FYP-II Phase**

Activities:

- Enhancement of ML model accuracy
- Real-time auto URL detection
- AI-based chatbot improvement
- Deployment on cloud server

Deliverables:

- Final system deployment
 - FYP-II final report and demo
-

Project Monitoring & Control

- Weekly supervisor meetings
- Progress tracking using milestones
- Issue resolution based on feedback
- Continuous documentation updates